Cypher Analytica – Data Analytics & Data Science Training Program

SQL Assignment: Sales Management System (100 Marks)

Below is the **enhanced full SQL Assignment** including:

- Basic to Advanced SQL concepts
- CTE (Common Table Expressions)
- UDF (User Defined Functions)
- Stored Procedures
- Window Functions

📄 SQL Assignment: Sales Management System (Advanced Version)

🎯 Objective

This assignment evaluates your SQL proficiency from fundamentals to advanced analytics:

- Data Retrieval
- Aggregations & Joins
- Subqueries
- CTEs, UDFs, Stored Procedures
- Analytical (Window) Functions

🗂️ Dataset

Use the following three tables (already provided in your CSV/SQL files):

- **Customers (CustomerID, CustomerName, City, Country)**
- **Products (ProductID, ProductName, Category, Price)**
- **Orders (OrderID, CustomerID, ProductID, OrderDate, Quantity)**

🧮 Assignment Tasks

Section A: Basic Queries (10 Marks)

1. Display all customers who are from India.
2. List all products under the *Accessories* category.
3. Show all orders placed in April 2024.
4. Display customer names and product names for all orders.
5. Show all products priced above ₹10,000 sorted by price descending.

Section B: Aggregation & Grouping (10 Marks)

6. Find the total quantity ordered by each customer.
7. Calculate total sales amount (Price × Quantity) for each order.
8. Find total revenue generated per product category.
9. Identify the customer who placed the **maximum total orders**.
10. Show average price of products in each category.

Section C: Joins (8 Marks)

11. Write a query to show CustomerName, ProductName, Quantity, and OrderDate.
12. Display a list of customers who purchased "Electronics" items only.
13. Find customers who have **not placed any orders**.
14. Display total sales for each customer along with their city and country.

Section D: Subqueries (8 Marks)

15. Display customers who have purchased products costing more than ₹25,000.
16. Find the product(s) whose price is greater than the **average price** of all products.
17. Show all orders where the quantity is **greater than the average quantity** ordered.
18. Display the product with the **highest total sales amount** using a subquery.

Section E: Common Table Expressions (CTE) (12 Marks)

19. Using a CTE, calculate total sales (Price × Quantity) for each customer.
20. Use a recursive CTE to generate a sequence of numbers from 1 to 10.
21. Using a CTE, find top 3 products by total revenue.

Section F: User Defined Functions (UDF) (10 Marks)

22. Create a UDF named TotalOrderAmount(OrderID) that returns total amount for that order.
23. Use this function to display OrderID and TotalAmount for all orders.

Section G: Stored Procedure (10 Marks)

24. Create a Stored Procedure named GetCustomerOrders(CustomerID) to list all orders placed by that customer.
25. Execute the stored procedure for CustomerID = 101.

Section H: Window Functions (16 Marks)

26. Show each customer's total purchase value and their rank based on total purchase (highest to lowest).
27. For each order, display the running total of quantity ordered by date.
28. Display product-wise average price and deviation of each product's price from its category average.
29. Display order details along with ROW_NUMBER() to show the sequence of orders by date.

Section I: Data Modification

30. Add a new customer. **(6 Marks)**
31. Update the price of "Pen Drive" to ₹1500.
32. Delete all orders placed before 2024-04-01.

Section J: Bonus Challenge (10 Marks)

33. Write a query to display the **Top 3 Customers by Total Revenue** using RANK() window function.
34. Create a stored procedure to return **monthly sales summary** (Month, Total_Orders, Total_Revenue).